

LAPORAN JOBSHEET
PEMROGRAMAN BERBASIS FRAMEWORK



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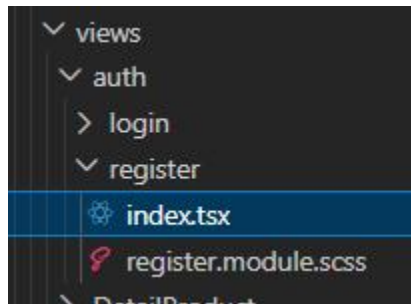
POLITEKNIK NEGERI MALANG TAHUN 2026

Github :

C. Langkah Praktikum

◆ Bagian 1 – Membuat Register View

- Buat folder pada views dengan nama register dan tambahkan 2 file yaitu index.tsx dan register.module.scss



- Modifikasi file index.tsx (pada folder views/auth/register/index.tsx)

```
14.praktikum-AuthRegister > my-app > src > views > auth > register > index.tsx > ...
1  import Link from "next/link";
2  import style from "../../auth/register/register.module.scss";
3
4
5  Windsurf: Refactor | Explain | Generate JSDoc | X
6  const TampilanRegister = () => {
7    return (
8      <div className={style.register}>
9        <h1 className={style.register__title}>Halaman Register</h1>
10       <Link href="/auth/login">Ke Halaman Login</Link>
11     </div>
12   );
13 };
14 export default TampilanRegister;
15
```

```
src > pages > views > auth > register > index.tsx > default
1  import Link from "next/link";
2  import style from "../../auth/register/register.module.scss";
3
4  Windsurf: Refactor | Explain | Generate JSDoc | X
5  const TampilanRegister = () => {
6    return (
7      <div className={style.register}>
8        <h1 className={style.register__title}>Halaman Register</h1>
9        <Link href="/auth/login">Ke Halaman Login</Link>
10      </div>
11    );
12  };
13  export default TampilanRegister;
```

- o Buka file register.tsx pada folder auth/register.tsx



- o Modifikasi file register.tsx (pada folder pages/auth/register.tsx)

```
14.praktikum-AuthRegister > my-app > src > pages > auth > register.tsx > ...
1  import Link from "next/link";
2  import TampilanRegister from "@/views/auth/register";
3
4  Windsurf: Refactor | Explain | Generate JSDoc | X
5  const halamanRegister = () => {
6    return (
7      <TampilanRegister />
8    );
9  };
10
11  export default halamanRegister;
12
```

```
src > pages > auth > register.tsx > default
1  import Link from "next/link";
2  import TampilanRegister from "../../views/auth/register";
3
4  const halamanRegister = () => {
5    return (
6      <TampilanRegister />
7    );
8  };
9
10
11  export default halamanRegister;
```

- o Modifikasi register.module.scss

```
14.praktikum-AuthRegister > my-app > src > views > auth > register > register.module.scss > .register_title
1  .register {
2    padding: 40px;
3  }
4
5  .register_title {
6    font-size: 28px;
7    font-weight: bold;
8  }
9
```

```
1  ✓ .register {
2    padding: 40px;
3  }
4
5  ✓ .register_title {
6    font-size: 28px;
7    font-weight: bold;
8  }
```

- Tambahkan form inputan pada file index.tsx (pada folder views/auth/register/index.tsx) Form berisi:
- Email

```
<div className={style.register__form__item}>
  <label
    htmlFor="email"
    className={style.register__form__item__label}
  >
    Email
  </label>
  <input
    type="email"
    id="email"
    name="email"
    placeholder="Email"
    className={style.register__form__item__input}
  />
</div>
```

```

<div className={style.register__form__item}>
  <label
    htmlFor="email"
    className={style.register__form__item__label}
  >
    Email
  </label>
  <input
    type="email"
    id="email"
    name="email"
    placeholder="Email"
    className={style.register__form__item__input}
  />
</div>

```

- Full Name

```

<div className={style.register__form__item}>
  <label
    htmlFor="Fullname"
    className={style.register__form__item__label}
  >
    Fullname
  </label>
  <input
    type="text"
    id="Fullname"
    name="Fullname"
    placeholder="Fullname"
    className={style.register__form__item__input}
  />
</div>

```

```

<div className={style.register__form__item}>
  <label
    htmlFor="Fullname"
    className={style.register__form__item__label}
  >
    Fullname
  </label>
  <input
    type="text"
    id="Fullname"
    name="Fullname"
    placeholder="Fullname"
    className={style.register__form__item__input}
  />
</div>

```

- Password

```

<div className={style.register__form__item}>
  <label
    htmlFor="Password"
    className={style.register__form__item__label}
  >
    Password
  </label>
  <input
    type="password"
    id="Password"
    name="Password"
    placeholder="Password"
    className={style.register__form__item__input}
  />
</div>

```

```

<div className={style.register__form__item}>
  <label
    htmlFor="Password"
    className={style.register__form__item__label}
  >
    Password
  </label>
  <input
    type="password"
    id="Password"
    name="Password"
    placeholder="Password"
    className={style.register__form__item__input}
  />
</div>

```

- Button Register

```

<button type="submit" className={style.register__form__item__button}>
  Register
</button>

```

```

<button type="submit" className={style.register__form__item__button}>
  Register
</button>

```

- Kode keseluruhannya sebagai berikut :

14.praktikum-AuthRegister > my-app > src > views > auth > register > index.tsx > TampilanRegister

```
1 import Link from "next/link";
2 import style from "../../auth/register/register.module.scss";
3
4 const TampilanRegister = () => {
5   return (
6     <div className={style.register}>
7       <h1 className={style.register__title}>Halaman Register</h1>
8       <div className={style.register__form}>
9         <form action="">
10           <div className={style.register__form__item}>
11             <label
12               htmlFor="email"
13               className={style.register__form__item__label}
14             >
15               Email
16             </label>
17             <input
18               type="email"
19               id="email"
20               name="email"
21               placeholder="Email"
22               className={style.register__form__item__input}
23             />
24           </div>
25           <div className={style.register__form__item}>
26             <label
27               htmlFor="Fullname"
28               className={style.register__form__item__label}
29             >
30               Fullname
31             </label>
32             <input
33               type="text"
34               id="Fullname"
35               name="Fullname"
36               placeholder="Fullname"
37               className={style.register__form__item__input}
38             />
39           </div>
40           <div className={style.register__form__item}>
41             <label
42               htmlFor="Password"
43               className={style.register__form__item__label}
44             >
45               Password
46             </label>
47             <input
48               type="password"
49               id="Password"
50               name="Password"
51               placeholder="Password"
52               className={style.register__form__item__input}
53             />
54           </div>
55           <button type="submit" className={style.register__form__item__button}>
56             Register
57           </button>
58         </form>
59         <br />
60         <p className={style.register__form__item__text}>
61           Sudah punya akun? <Link href="/auth/login">Ke Halaman Login</Link>
62         </p>
63       </div>
64     </div>
65   );
66 }
67
68 export default TampilanRegister;
```



```

1 import Link from "next/link";
2 import style from "../register.module.scss";
3
4 Windsurf: Refactor | Explain | Generate JSDoc | X
5 const TampilanRegister = () => {
6   return (
7     <div className={style.register}>
8       <h1 className={style.register_title}>Halaman Register</h1>
9       <div className={style.register_form}>
10         <form action="">
11           <div className={style.register_form_item}>
12             <label
13               htmlFor="email"
14               className={style.register_form_item_label}
15             >
16               Email
17             </label>
18             <input
19               type="email"
20               id="email"
21               name="email"
22               placeholder="Email"
23               className={style.register_form_item_input}
24             />
25           </div>
26           <div className={style.register_form_item}>
27             <label
28               htmlFor="Fullname"
29               className={style.register_form_item_label}
30             >
31               Fullname
32             </label>
33             <input
34               type="text"
35               id="Fullname"
36               name="Fullname"
37               placeholder="Fullname"
38               className={style.register_form_item_input}
39             />
40           </div>
41           <div className={style.register_form_item}>
42             <label
43               htmlFor="Password"
44               className={style.register_form_item_label}
45             >
46               Password
47             </label>
48             <input
49               type="password"
50               id="Password"
51               name="Password"
52               placeholder="Password"
53               className={style.register_form_item_input}
54             />
55           </div>
56           <button type="submit" className={style.register_form_item_button}>
57             Register
58           </button>
59         </form>
60         <br />
61         <p className={style.register_form_item_text}>
62           Sudah punya akun? <Link href="/auth/login">Ke Halaman Login</Link>
63         </p>
64       </div>
65     </div>
66   );
67 };
68
69 export default TampilanRegister;
70

```

- Modifikasi register.module.scss

14.praktikum-AuthRegister > my-app > src > views > auth > register > register.module.scss > .register > &__form

```
1  .register {
2    min-height: 100vh;
3    display: flex;
4    flex-direction: column; /* ← penting */
5    align-items: center;
6    justify-content: center;
7    background: linear-gradient(135deg, #1e3c72, #2a5298);
8    padding: 20px;
9
10   &__title {
11     width: 100%;
12     text-align: center;
13     font-size: 36px;
14     font-weight: 700;
15     color: #ffffff;
16     margin-bottom: 40px;
17   }
18
19   &__form {
20     background: #ffffff;
21     width: 100%;
22     max-width: 420px;
23     padding: 40px;
24     border-radius: 16px;
25     box-shadow: 0 20px 40px rgba(0, 0, 0, 0.15);
26
27     &__item {
28       display: flex;
29       flex-direction: column;
30       margin-bottom: 20px;
31
32       &__label {
33         font-size: 14px;
34         font-weight: 600;
35         margin-bottom: 8px;
36         color: #333;
37       }
38
39       &__input {
40         padding: 12px 14px;
41         border-radius: 10px;
42         border: 1px solid #ddd;
43         font-size: 14px;
44         transition: all 0.3s ease;
45
46         &:focus {
47           outline: none;
48           border-color: #2a5298;
49           box-shadow: 0 0 3px rgba(42, 82, 152, 0.2);
50         }
51       }
52     }
53   }
```

```

54     button {
55         width: 100%;
56         padding: 12px;
57         border-radius: 10px;
58         border: none;
59         background: #2a5298;
60         color: #ffffff;
61         font-weight: 600;
62         cursor: pointer;
63         transition: 0.3s ease;
64     }
65     &:hover {
66         background: #1e3c72;
67         transform: translateY(-2px);
68     }
69     }
70 }
71 }
72
73 /* Animation */
74 @keyframes fadeIn {
75     from {
76         opacity: 0;
77         transform: translateY(15px);
78     }
79     to {
80         opacity: 1;
81         transform: translateY(0);
82     }
83 }
84
85 /* Responsive */
86 @media (max-width: 768px) {
87     .register {
88         &_form {
89             padding: 30px 20px;
90         }
91     }
92     &_title {
93         font-size: 26px;
94     }
95 }
96 }
97

```

```

1 .register {
2   min-height: 100vh;
3   display: flex;
4   flex-direction: column; /* <- penting */
5   align-items: center;
6   justify-content: center;
7   background: linear-gradient(135deg, #1e3c72, #2a5298);
8   padding: 20px;
9
10  &_title {
11    width: 100%;
12    text-align: center;
13    font-size: 36px;
14    font-weight: 700;
15    color: #ffffff;
16    margin-bottom: 40px;
17  }
18
19  &_form {
20    background: #ffffff;
21    width: 100%;
22    max-width: 420px;
23    padding: 40px;
24    border-radius: 16px;
25    box-shadow: 0 20px 40px rgba(0, 0, 0, 0.15);
26
27    &_item {
28      display: flex;
29      flex-direction: column;
30      margin-bottom: 20px;
31
32      &_label {
33        font-size: 14px;
34        font-weight: 600;
35        margin-bottom: 8px;
36        color: #333;
37      }
38
39      &_input {
40        padding: 12px 14px;
41        border-radius: 10px;
42        border: 1px solid #ddd;
43        font-size: 14px;
44        transition: all 0.3s ease;
45
46        &:focus {
47          outline: none;
48          border-color: #2a5298;
49          box-shadow: 0 0 3px rgba(42, 82, 152, 0.2);
50        }
51      }
52    }
53  }
54 }

```

```

54 }
55 button {
56     width: 100%;
57     padding: 12px;
58     border-radius: 10px;
59     border: none;
60     background: #2a5298;
61     color: #ffffff;
62     font-weight: 600;
63     cursor: pointer;
64     transition: 0.3s ease;
65
66     &:hover {
67         background: #1e3c72;
68         transform: translateY(-2px);
69     }
70 }
71
72 /* Animation */
73 @keyframes fadeIn {
74     from {
75         opacity: 0;
76         transform: translateY(15px);
77     }
78     to {
79         opacity: 1;
80         transform: translateY(0);
81     }
82 }
83
84 /* Responsive */
85 @media (max-width: 768px) {
86     .register {
87         &__form {
88             padding: 30px 20px;
89         }
90
91         &__title {
92             font-size: 26px;
93         }
94     }
95 }

```

- Jalankan browsernya <http://localhost:3000/auth/register> sehingga tampilan sebagai berikut

Halaman Register

Email

Fullname

Password

Register

Sudah punya akun? Ke Halaman Login

Halaman Register

Email

Fullname

Password

Register

Sudah punya akun? [Ke Halaman Login](#)

◆ Bagian 2 – Membuat API Register

- Buka file servicefirebase.ts pada folder src/utils/db dan modifikasi

14.praktikum-AuthRegister > my-app > src > utils > db > TS servicefirebase.ts > ...

```
1  import {
2    getFirestore,
3    collection,
4    getDocs,
5    Firestore,
6    getDoc,
7    doc,
8    query,
9    addDoc,
10   where,
11 } from "firebase/firestore";
12 import app from "../firebase";
13
14 const db = getFirestore(app);
15
16 Windsurf: Refactor | Explain | Generate JSDoc | X
17 export async function retrieveProducts(collectionName: string) {
18   const snapshot = await getDocs(collection(db, collectionName));
19   const data = snapshot.docs.map((doc) => ({
20     id: doc.id,
21     ...doc.data(),
22   }));
23   return data;
24 }
25
26 Windsurf: Refactor | Explain | Generate JSDoc | X
27 export async function retrieveDataByID(collectionName: string, id: string) {
28   const snapshot = await getDoc(doc(db, collectionName, id));
29   const data = snapshot.data();
30   return data;
31 }
32
33 Windsurf: Refactor | Explain | Generate JSDoc | X
34 export async function signUp(
35   userData: {
36     email: string;
37     fullname: string;
38     password: string;
39   },
40   callback: Function,
41 ) {
42   const q = query(
43     collection(db, "users"),
44     where("email", "==", userData.email),
45   );
46   const querySnapshot = await getDocs(q);
47   const data = querySnapshot.docs.map((doc) => ({
48     id: doc.id,
49     ...doc.data(),
50   }));
51   // console.log("Query result:", data);
52 }
```

```

50     if (data.length > 0) {
51         // user belum ada → boleh daftar
52         // await addDoc(collection(db, "users"), userData);
53         // console.log("User registered:", data);
54         callback({
55             status: "success",
56             message: "User registered successfully",
57         });
58     } else {
59         callback({
60             status: "error",
61             message: "User already exists",
62         });
63     }
64 }
65

```

```

1  import {
2      getFirestore,
3      collection,
4      getDocs,
5      Firestore,
6      getDoc,
7      doc,
8      query,
9      addDoc,
10     where,
11 } from "firebase/firestore";
12 import app from "../firebase";
13
14 const db = getFirestore(app);
15
16 Windsurf: Refactor | Explain | Generate JSDoc | ✕
17 export async function retrieveProducts(collectionName: string) {
18     const snapshot = await getDocs(collection(db, collectionName));
19     const data = snapshot.docs.map((doc) => ({
20         id: doc.id,
21         ...doc.data(),
22     }));
23     return data;
24 }
25
26 Windsurf: Refactor | Explain | Generate JSDoc | ✕
27 export async function retrieveDataByID(collectionName: string, id: string) {
28     const snapshot = await getDoc(doc(db, collectionName, id));
29     const data = snapshot.data();
30     return data;
31 }
32

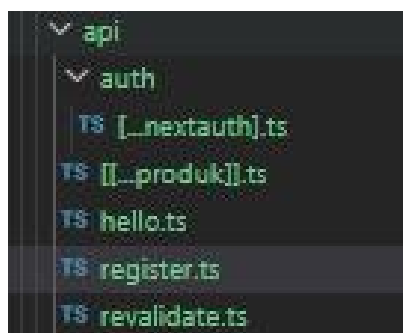
```

```

29 }
30
Windsurf: Refactor | Explain | Generate JSDoc | ×
31 export async function signUp(
32   userData: {
33     email: string;
34     fullname: string;
35     password: string;
36   },
37   callback: Function,
38 ) {
39   const q = query(
40     collection(db, "users"),
41     where("email", "=", userData.email),
42   );
43   const querySnapshot = await getDocs(q);
44   const data = querySnapshot.docs.map((doc) => ({
45     id: doc.id,
46     ...doc.data(),
47   }));
48   // console.log("Query result:", data);
49   if (data.length > 0) {
50     // user belum ada -> boleh daftar
51     // await addDoc(collection(db, "users"), userData);
52     // console.log("User registered:", data);
53     callback({
54       status: "success",
55       message: "User registered successfully",
56     });
57   } else {
58     callback({
59       status: "error",
60       message: "User already exists",
61     });
62   }
63 }

```

- Buat file register.ts pada folder api



- Modifikasi file register.ts

14.praktikum-AuthRegister > my-app > src > pages > api > TS register.ts > handler

```
1 // Next.js API route support: https://nextjs.org/docs/api-routes/introduction
2 import { signUp } from '@utils/db/servicefirebase'
3 import type { NextApiRequest, NextApiResponse } from 'next'
4
5 type Data = {
6   name: string
7   alamat: string
8 }
9
10 export default async function handler(
11   req: NextApiRequest,
12   res: NextApiResponse<Data>
13 ) {
14   if(req.method === "POST"){
15     await signUp(req.body, (result: {status: string, message: string}) => {
16       if(result.status === "success"){
17         res.status(200).json({ name: result.message, alamat: "" });
18       } else {
19         res.status(400).json({ name: result.message, alamat: "" });
20       }
21     });
22   }
23   else {
24     res.status(405).json({ name: "Method not allowed", alamat: "" });
25   }
26 }
```

Windsurf: Refactor | Explain

Windsurf: Refactor | Explain | Generate JSDoc | X

src > pages > api > TS register.ts > handler

```
1 // Next.js API route support: https://nextjs.org/docs/api-routes/introduction
2 import { signUp } from '@utils/db/servicefirebase'
3 import type { NextApiRequest, NextApiResponse } from 'next'
4
5 type Data = {
6   name: string
7   alamat: string
8 }
9
10 export default async function handler(
11   req: NextApiRequest,
12   res: NextApiResponse<Data>
13 ) {
14   if(req.method === "POST"){
15     await signUp(req.body, (result: {status: string, message: string}) => {
16       if(result.status === "success"){
17         res.status(200).json({ name: result.message, alamat: "" });
18       } else {
19         res.status(400).json({ name: result.message, alamat: "" });
20       }
21     });
22   }
23   else {
24     res.status(405).json({ name: "Method not allowed", alamat: "" });
25   }
26 }
```

Windsurf: Refactor | Explain

Windsurf: Refactor | Explain | Generate JSDoc | X

-

- Modifikasi index.tsx pada folder register (tambahkan beberapa code)

14.praktikum-AuthRegister > my-app > src > views > auth > register > index.tsx > TampilanRegister > handleSubmit

```
1 import Link from "next/link";
2 import style from "../../auth/register/register.module.scss";
3 import { useState } from "react";
4 import { useRouter } from "next/router";
5
6 Windsurf: Refactor | Explain | Generate JSDoc | X
7 const TampilanRegister = () => {
8   const [isLoading, setIsLoading] = useState(false);
9   const { push } = useRouter();
10  const [error, setError] = useState("");
11  Windsurf: Refactor | Explain | Generate JSDoc | X
12  const handleSubmit = async (event: React.FormEvent<HTMLFormElement>) => {
13    event.preventDefault();
14    const form = event.currentTarget;
15    const formData = new FormData(event.currentTarget);
16    const email = formData.get("email") as string;
17    const fullname = formData.get("Fullname") as string;
18    const password = formData.get("Password") as string;
19    const response = await fetch("/api/register", {
20      method: "POST",
21      headers: {
22        "Content-Type": "application/json",
23      },
24      body: JSON.stringify({ email, fullname, password }),
25    });
26    // const result = await response.json();
27    // console.log(result);
28    if (response.status === 200) {
29      form.reset();
30      // event.currentTarget.reset();
31      setIsLoading(false);
32      push("/auth/login");
33    } else {
34      setIsLoading(false);
35      setError(
36        response.status === 400 ? "User already exists" : "An error occurred",
37      );
38    }
39  };
40
41  return (
42    <div className={style.register}>
43      <h1 className={style.register_title}>Halaman Register</h1>
44      <div className={style.register_form}>
45        <form onSubmit={handleSubmit}>
```



```

1 import Link from "next/link";
2 import style from "../register.module.scss";
3 import { useState } from "react";
4 import { useRouter } from "next/router";
5
6 Windsurf: Refactor | Explain | Generate JSDoc | X
7 const TampilanRegister = () => {
8   const [isLoading, setIsLoading] = useState(false);
9   const { push } = useRouter();
10   const [error, setError] = useState("");
11
12   Windsurf: Refactor | Explain | Generate JSDoc | X
13   const handleSubmit = async (event: React.FormEvent<HTMLFormElement>) => {
14     event.preventDefault();
15     setIsLoading(true);
16     setError("");
17
18     const form = event.currentTarget;
19     const formData = new FormData(form);
20
21     const email = formData.get("email") as string;
22     const fullName = formData.get("Fullname") as string;
23     const password = formData.get("Password") as string;
24
25     const response = await fetch("/api/register", {
26       method: "POST",
27       headers: {
28         "Content-Type": "application/json",
29       },
30       body: JSON.stringify({ email, fullName, password }),
31     });
32
33     if (response.status === 200) {
34       form.reset();
35       setIsLoading(false);
36       push("/auth/login");
37     } else {
38       setIsLoading(false);
39       setError(
40         response.status === 400
41         ? "User already exists"
42         : "An error occurred. Please try again later."
43       );
44     }
45   };
46
47   return (
48     <div className={style.register}>
49       <h1 className={style.register__title}>Halaman Register</h1>
50
51       <div className={style.register__form}>
52         <form onSubmit={handleSubmit}>
53           <div className={style.register__form_item}>
54             <label className={style.register__form_item_label}>

```

- Buka browser <http://localhost:3000/auth/register> isikan data dan klik register. Jika berhasil maka akan masuk ke menu login

Halaman Register

Email

Fullname

Password

Register

Sudah punya akun? Ke Halaman Login

Halaman Register

Email

yandaffa19@gmail.com

Fullname

yan daffa

Password

.....

Register

Sudah punya akun? Ke Halaman Login

◆ Bagian 3 – Install bcrypt

- `npm install bcrypt --force`

```
PS D:\1.MateriKuliah\2026-ganjil\PemogramanFramework\Code\14.praktikum-AuthRegister\my-app> npm install bcrypt --force
npm warn using --force Recommended protections disabled.
```

```
added 3 packages, and audited 447 packages in 5s
```

```
146 packages are looking for funding
  run `npm fund` for details
```

```
found 0 vulnerabilities
```



```
D:\tugas\semester 6\framework\js15\code\my-app>npm install bcrypt --force
npm warn using --force Recommended protections disabled.
```

```
added 3 packages, and audited 490 packages in 21s
```

```
158 packages are looking for funding
  run `npm fund` for details
```

```
1 moderate severity vulnerability
```

```
To address all issues, run:
  npm audit fix --force
```

```
Run `npm audit` for details.
```

- npm install --save-dev @types/bcrypt --force

```
PS D:\1.MateriKuliah\2026-ganjil\PemogramanFramework\Code\14.praktikum-AuthRegister\my-app> npm install --save-dev @types/bcrypt --force
npm warn using --force Recommended protections disabled.
```

```
added 1 package, and audited 448 packages in 3s
```

```
146 packages are looking for funding
  run `npm fund` for details
```

```
found 0 vulnerabilities
```

```
D:\tugas\semester 6\framework\js15\code\my-app>npm install --save-d
ev @types/bcrypt --force
npm warn using --force Recommended protections disabled.
```

```
added 1 package, and audited 491 packages in 4s
```

```
158 packages are looking for funding
  run `npm fund` for details
```

```
1 moderate severity vulnerability
```

```
To address all issues, run:
  npm audit fix --force
```

```
Run `npm audit` for details.
```

- Buka file servicefirebase.ts pada folder src/utils/db dan modifikasi

14.praktikum-AuthRegister > my-app > src > utils > db > TS servicefirebase.ts > ...

```
1  import {
2    getFirestore, collection,
3    getDocs, Firestore, getDoc, doc,
4    query, addDoc, where,
5  } from "firebase/firestore";
6  import app from "firebase";
7  import bcrypt from "bcrypt";
8
9  const db = getFirestore(app);
10
11  Windsurf Refactor | Explain | Generate JSDoc | X
12  > export async function retrieveProducts(collectionName: string) { ...
13  }
14
15  Windsurf Refactor | Explain | Generate JSDoc | X
16  > export async function retrieveDataByID(collectionName: string, id: string) { ...
17  }
18
19  Windsurf Refactor | Explain | Generate JSDoc | X
20  export async function signUp(
21    userData: {
22      email: string;
23      fullname: string;
24      password: string;
25      role?: string;
26    },
27    callback: Function,
28  ) {
29    const q = query( ...
30    );
31    const querySnapshot = await getDocs(q);
32    const data = querySnapshot.docs.map((doc) => ({ ...
33    }));
34    // console.log("Query result:", data);
35
36    if (data.length > 0) {
37      // user belum ada + boleh daftar
38      // await addDoc(collection(db, "users"), userData);
39      // console.log("User registered:", data);
40      callback({
41        status: "success",
42        message: "User registered successfully",
43      });
44    } else {
45      userData.password = await bcrypt.hash(userData.password, 10);
46      userData.role = "user";
47      await addDoc(collection(db, "users"), userData)
48        .then(() => {
49          callback({
50            status: "success",
51            message: "User registered successfully",
52          });
53        })
54        .catch((error) => {
55          callback({
56            status: "error",
57            message: error.message,
58          });
59        });
60    }
61  }
62  }
```

```

src > utils > db > ts servicefirebase.ts > ...
1 import {
2   getFirestore, collection, getDocs, Firestore, getDoc, doc, query, addDoc,
3   where,
4 } from "firebase/firestore";
5 import app from "../firebase";
6 import bcrypt from "bcrypt";
7
8 const db = getFirestore(app);
9
10 export async function retrieveProducts(collectionName: string) {
11   const snapshot = await getDocs(collection(db, collectionName));
12   const data = snapshot.docs.map((doc) => ({
13     id: doc.id,
14     ...doc.data(),
15   }));
16   return data;
17 }
18
19 export async function retrieveDataByID(collectionName: string, id: string) {
20   const doc = await getDoc(doc(db, collectionName, id));
21   return doc.data();
22 }
23
24 export async function signUp(
25   userData: {
26     email: string;
27     fullname: string;
28     password: string;
29     role?: string;
30   },
31   callback: Function,
32 ) {
33   const q = query(
34     collection(db, "users"),
35     where("email", "==", userData.email),
36   );
37   const querySnapshot = await getDocs(q);
38   const data = querySnapshot.docs.map((doc) => ({
39     id: doc.id,
40     ...doc.data(),
41   }));
42   // console.log("Query result:", data);
43   if (data.length > 0) {
44     // user belum ada -> boleh daftar
45     // await addDoc(collection(db, "users"), userData);
46     // console.log("User registered:", data);
47     callback({
48       status: "success",
49       message: "User registered successfully",
50     });
51   } else {
52     userData.password = await bcrypt.hash(userData.password, 10);
53     userData.role = "user";
54     await addDoc(collection(db, "users"), userData)
55       .then(() => {
56         callback({
57           status: "success",
58           message: "User registered successfully",
59         });
60       })
61       .catch((error) => {
62         callback({
63           status: "error",
64           message: error.message,
65         });
66       });
67   }
68 }
69

```

- Jalankan browser <http://localhost:3000/auth/register> dan input data setelah itu klik register

Halaman Register

Email

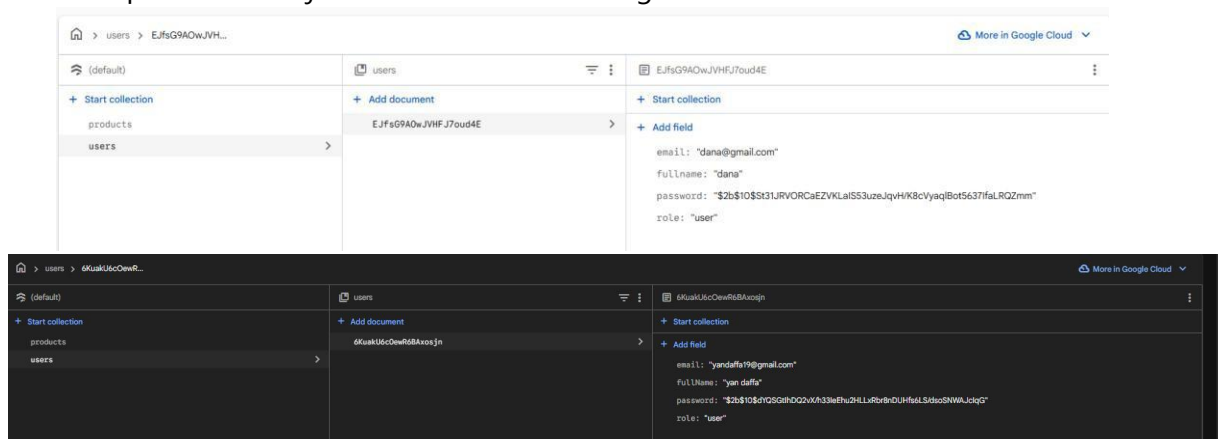
Fullname

Password

[Register](#)

Sudah punya akun? [Ke Halaman Login](#)

- Buka pada firebase jika berhasil maka data register akan masuk



- Jika user memasukkan data yang sama sistem tidak akan memproses tetapi permasalahannya user memasukkan data yang sama tidak ada pemberitahuan pada layar maka dari itu perlu ada perubahan pada code index.tsx pada folder views/auth/register

- o Line 41

```

39   return (
40     <div className={style.register}>
41       {error && <p className={style.register__error}>{error}</p>}
42       <h1 className={style.register__title}>Halaman Register</h1>

```

```

return (
  <div className={style.register}>
    {error && <p className={style.register__error}>{error}</p>}
    <h1 className={style.register__title}>Halaman Register</h1>

```

- o Line 94 dan 96

```

91   <button
92     type="submit"
93     className={style.register__form__item__button}
94     disabled={isLoading}
95   >
96     {isLoading ? "Loading..." : "Register"}
97   </button>

```

```

<button
  type="submit"
  disabled={isLoading}
  className={style.register_form_item_button}
>
  {isLoading ? "Loading..." : "Register"}
</button>

```

- Line 34 rubah menjadi email

```

14.praktikum-AuthRegister > my-app > src > views > auth > register > index.tsx > TampilanRegister > handleSubmit > response
6   const TampilanRegister = () => {
10    const handleSubmit = async (event: React.FormEvent<HTMLFormElement>) => {
26      if (response.status === 200) {
27        form.reset();
28        // event.currentTarget.reset();
29        setIsLoading(false);
30        push("/auth/login");
31      } else {
32        setIsLoading(false);
33        setError(
34          response.status === 400 ? "Email already exists" : "An error occurred",
35        );
36      }
37    };

```

```

    if (response.status === 200) {
      form.reset();
      setIsLoading(false);
      push("/auth/login");
    } else {
      setIsLoading(false);
      setError(
        response.status === 400
          ? "Email already exists"
          : "An error occurred."
      );
    }
  };

```

- Modifikasi juga pada register.module.scss

```

14.praktikum-AuthRegister > my-app > src > views > auth > register > register.module.scss > .register > &__error
1   .register {
2     min-height: 100vh;
3     display: flex;
4     flex-direction: column; /* ← penting */
5     align-items: center;
6     justify-content: center;
7     background: linear-gradient(135deg, #1e3c72, #2a5298);
8     padding: 20px;
9
10    &__title { ...
17  }
18
19    &__form { ...
20  }
21
22    &__error {
23      color: red;
24      font-size: 14px;
25      margin-top: 8px;
26    }
27  }

```

```

26
27 ✓
28 &__error {
29   color : red;
30   font-size: 14px;
31   margin-top: 8px;
32 }

```

- Jika berhasil maka hasilnya seperti berikut

Email already exists

Halaman Register

Email

dana@gmail.com

Fullname

dana

Password

....

Register

Sudah punya akun? Ke Halaman Login

Email already exists

Halaman Register

Email

yandaffa19@gmail.com

Fullname

asd

Password

..+

Register

Sudah punya akun? Ke Halaman Login

- Tambahkan loading dengan menambahkan kode pada index.tsx

```
14.praktikum-AuthRegister > my-app > src > views > auth > register > index.tsx > TampilanRegister > handleSubmit
1  import Link from "next/link";
2  import style from "../../auth/register/register.module.scss";
3  import { useState } from "react";
4  import { useRouter } from "next/router";
5
6  Windsurf: Refactor | Explain | Generate JSDoc | X
7  const TampilanRegister = () => {
8    const [isLoading, setIsLoading] = useState(false);
9    const { push } = useRouter();
10   const [error, setError] = useState("");
11   const handleSubmit = async (event: React.FormEvent<HTMLFormElement>) => {
12     setError("");
13     setIsLoading(true);
14     event.preventDefault();
```

```
Windsurf: Refactor | Explain | Generate JSDoc | X
11 const handleSubmit = async (event: React.FormEvent<HTMLFormElement>) => {
12   setError("");
13   setIsLoading(true);
14   event.preventDefault();
```

- Jika berhasil maka hasilnya akan muncul loading saat klik register

✎ D. Pengujian

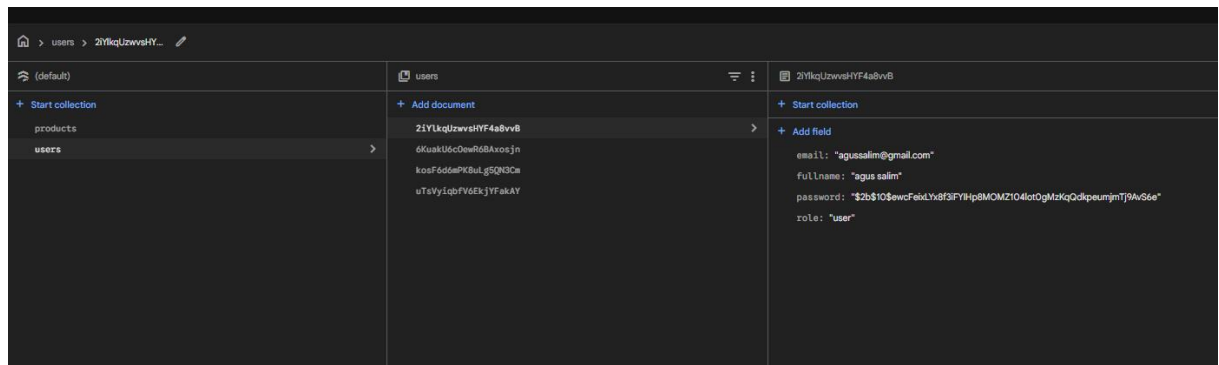
✎ Uji 1 – Register Baru

Input:

- Email baru

Hasil:

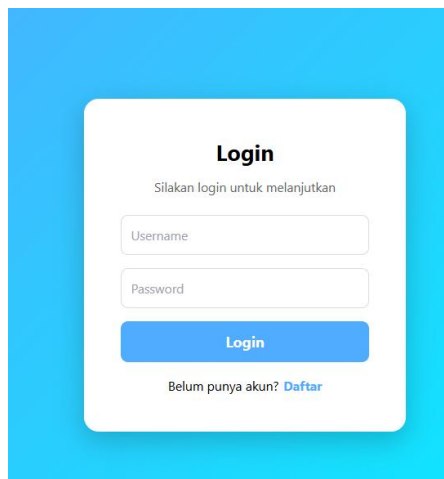
- Data tersimpan di Firestore



- Password ter-hash

```
password: "$2b$10$ewcFeiLYx8f3iFYIHp8MOMZ104lot0gMzKqQdkpeumjmTj9AvS6e"
```

- Redirect ke login



➤ Uji 2 – Email Sudah Ada

Input:

- Email yang sama

Hasil:

- Error 400
- Message: Email already exists

Email already exists

Halaman Register

Email

Fullname

Password

Register

Sudah punya akun? Ke Halaman Login

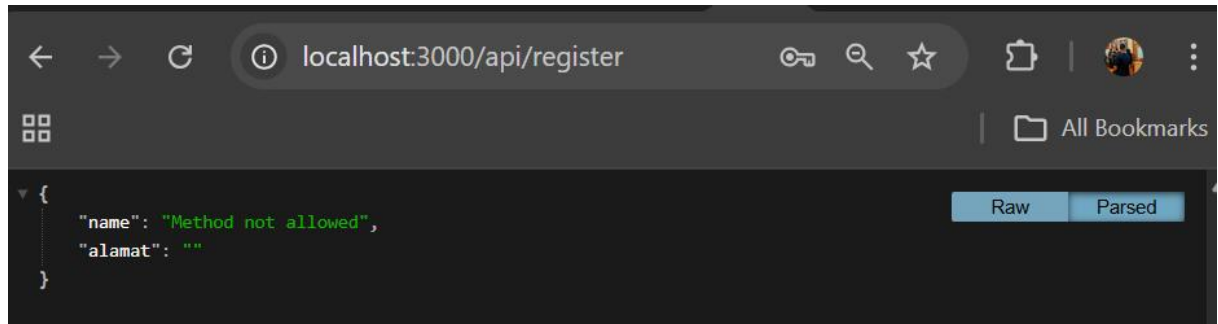
Uji 3 – Method GET

Akses:

/api/register

Hasil:

- 405 Method Not Allowed



#E. Struktur Database (Firestore)

Collection: users

Field:

Field	Tipe
fullName	string
email	string
password	string (hashed)
role	string
createdAt	timestamp

> G. Tugas Praktikum

1. Implementasikan register terhubung database.
2. Tambahkan validasi:
 - Email wajib
 - Password minimal 6 karakter

```

) {
  try {
    // ===== VALIDASI =====
    if (!userData.email) {
      return callback({
        status: "error",
        message: "Email wajib diisi",
      });
    }

    if (!userData.password) {
      return callback({
        status: "error",
        message: "Password wajib diisi",
      });
    }

    if (userData.password.length < 6) {
      return callback({
        status: "error",
        message: "Password minimal 6 karakter",
      });
    }
  }
}

```

3. Tambahkan role default "member".

```

await addDoc(collection(db, "users"), {
  email: userData.email,
  fullname: userData.fullname,
  password: hashedPassword,
  role: "member",
});

```

4. Tampilkan pesan error di UI.

Email wajib diisi

Halaman Register

Email

Fullname

Password

Register

Sudah punya akun? [Ke Halaman Login](#)



H. Pertanyaan Analisis

1. Mengapa password harus di-hash?

Agar data yang disimpan di database bisa lebih aman, diakrenakan jika tidak di hash maka password akan bisa diakses oleh pihak yang bisa mengakses database yang memungkinkan untuk disalahgunakan

2. Apa perbedaan addDoc dan setDoc?

addDoc digunakan ketika ingin menggenerate ID dokumen untuk data baru secara otomatis di firebase, sedangkan setDoc digunakan untuk menentukan ID dokumen secara manual

3. Mengapa perlu validasi method POST?

Validasi method POST diperlukan untuk memastikan agar API hanya bisa menerima request POST

jika tidak nantinya akan menimbulkan security issue pada aplikasi

4. Apa risiko jika email tidak dicek unik?

Resikonya adalah user dapat mendaftarkan akun baru dengan email yang sama yang menyebabkan adanya duplikasi email pada database yang nantinya bisa menyebabkan sistem bermasalah untuk menentukan akun mana yang harus diakses, dan bisa jadi nantinya bisa menimbulkan pencurian akun

5. Apa fungsi role pada user?

Fungsi role nantinya dapat digunakan sebagai pembeda atau pengelompokan terhadap fitur-fitur yang bisa diakses pada setiap rolennya

